SMSE AI Toolkit: Material Class and Properties Prediction

A Thesis

Submitted *by*

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DECLARATION

"I hereby declare that this submission is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text."

Place**: NIT CALICUT**

Date: **May 1st, 2023**

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CERTIFICATE

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ABSTRACT

This paper presents an integrated framework that combines image recognition, steel property prediction, and citation generation to facilitate efficient analysis and information retrieval for diverse applications. The framework encompasses the recognition of micrographs of fibers, biological samples, film-coated surfaces, and porous sponges, as well as the prediction of steel properties based on the number of other atoms present inthe steel formula. Additionally, the framework incorporates an API-based citation generator, enabling the automatic generation of accurate citations for references.

The image recognition component of the framework employs advanced machine learning algorithms to analyze and classify micrographs. It leverages deep neural networks trained on vast datasets, enabling accurate identification and characterization of fibers, biological specimens, film-coated surfaces, and porous sponges. This capability facilitates faster analysis and classification of complex materials and structures, enhancing the efficiency of scientific research and industrial applications.

The steel property prediction component utilizes statistical models to predict the properties of steel based on the presence and composition of other atoms in the steel formula. By considering the elemental composition and their interactions, the framework provides insights into the behavior and performance of various steel alloys. This predictive capability aids in designing and optimizing steel compositions for desired properties, leading to improved material performance in structural and engineering applications.

Furthermore, the framework incorporates a citation generator module that utilizes an API to automatically generate accurate citations for academic references. By integrating with established citation databases, the system ensures adherence to citation standards and streamlines the process of creating bibliographies and reference lists. This feature provides researchers and authors with a convenient and time-saving solution for maintaining proper citation practices.

The complete framework is deployed on a website, offering a user-friendly interface for accessing its functionalities. Users can upload micrograph images for recognition, input steel formula details for property prediction, and generate citations for their references. The website provides an intuitive platform for researchers, scientists, and engineers to leverage these capabilities and obtain valuable insights efficiently.

Nomenclature

Keywords

Image Classification & Recognition, SEM & TEM images, Linear Regression, Data Explosion, Batch Size, Epochs, Data Mining

Abbreviations:

1- CNN - Convolutional Neural Network
2- ANN - Artificial Neural Network
3- ML- Machine Learning
4- SEM- Scanning Electron Microscopy
5- TEM- Transmission electron microscopy
6- EDA- Exploratory Data Analysis
7- UI- User Interface
8- UX- User Experience
9- ReLU - Rectified Linear Unit

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Chapter 1

1.1 Introduction

Traditional machine learning methods, such as multilayer perceptron machines and support vector machines, often utilize shallow architectures to handle a limited number of samples and computational units. However, when dealing with complex classification tasks that involve target objects with diverse meanings, these approaches exhibit inadequate performance and generalization capacity.

One widely adopted deep learning architecture, known as the convolutional neural network (CNN), has become a staple in image processing. In recent years, CNN- based representations have been extensively used due to their ability to provide spatial and spectral contextual sensitivity, which is crucial for accurate pixel classification. This is achieved by amalgamating a variety of discriminant appearance components.

This study aims to address the challenging task of VFSR (Visual Full-Scale Range) picture categorization efficiently. Ensuring the safe operation of nuclear power plants necessitates periodic inspection of their components.

In the realm of crack detection on metal surfaces, several vision-based techniques have been developed. However, detecting these cracks is challenging due to their small size and the presence of noise patterns on the component surfaces. Predicting visual attention data from various types of media is valuable for content producers and can be used to drive coding algorithms effectively.

Our current endeavor involves leveraging CNNs, ANNs (Artificial Neural Networks), databases, and website hosting to develop an AI Toolkit for the School of Materials Science and Engineering at NITC (National Institute of Technology Calicut). This toolkit aims to assist with educational and research purposes, offering applications such as micrograph recognition for analyzing micrograph images, a property prediction toolkit, and various other mini tools to enhance utility.

It's important to note that this project is open source, meaning it can be used and modified by anyone interested, without any copyright infringement concerns.

1.2 Literature Review and Gap:

(i) The SMSE AI Toolkit is a collection of mini-projects designed to enhance materials science and engineering research through the application of artificial intelligence techniques. Among the mini-projects included in the toolkit is an image recognition model, which was trained using 1000 micrographs to distinguish between porous sponge, biological, and fiber materials. The model utilizes the MaxPooling2D layer from the Keras deep learning library to extract the most relevant features from the images, and achieves an accuracy of 92% in classification.

(ii) Another mini-project included in the SMSE AI Toolkit is a steel recognition model, which uses multiple linear regression analysis to predict mechanical properties such as tensile strength, proof strength, and elongation based on the constituent elements in steel alloys. This model was inspired by the work of Ford et al. (2021),

who developed a machine learning-based approach to predict properties of two-phase materials using microstructural descriptors and finite element analysis.

(iii) The development of the steel recognition model also drew from previous research in the field, including the comprehensive mineral database compiled by Venugopal (2021) and the SEM dataset produced by Aversa et al. (2018). Additionally, the Materials Project, a materials genome approach to accelerating materials innovation described by Jain et al. (2013), was utilized as a resource for materials data and analysis.

Through the incorporation of these various resources and techniques, the SMSE AI Toolkit offers a powerful suite of tools for materials science and engineering researchers. The toolkit's citation generator and database lookup tool further facilitate the research process, while the website hosting all of the mini-projects provides a convenient and accessible platform for utilization of the toolkit.

1.2.1 Literature Gap-

Based on the literature reviewed, a significant gap in the research is the lack of attention given to the development of a comprehensive AI toolkit that can assist in the prediction and classification of both microstructures and mechanical properties of materials.

Although there have been previous studies on image recognition and steel recognition models, very few have combined both fields to develop a comprehensive toolkit that can provide a broad range of predictions for different materials.

The SMSE AI Toolkit developed in this project fills this gap by providing multiple mini projects that can be used to pre33dict both microstructures and mechanical properties, thus serving as a valuable tool for materials science researchers and engineers. Further research is needed to explore the full potential of such a toolkit and its potential applications in the industry.

Chapter 2

Image Recognition Model

2.1 What is ANN (Artificial Neural Networks)

An ANN (Artificial Neural Network or Neuron Network) is a computational model that simulates the function of nerve cells in the human brain.
Artificial neural networks (ANNs) employ learning algorithms that can autonomously make modifications - or learn - when fresh input is received. As a result, they are a very useful tool for non-linear statistical data modelling.
An artificial neural network has three or more layers that are interconnected. The first layer consists of input neurons. Those neurons send data on to the deeper layers, which in turn will send the final output data to the last output layer.
All the inner layers are hidden and are formed by units which adaptively change the information received from layer to layer through a series of transformations. Each layer acts both as an input and output layer that allows the ANN to understand more complex objects. Collectively, these inner layers are called the neural layer.
The neural layer units attempt to learn about the information acquired by weighing it in accordance with the ANN's internal framework. These rules enable units to produce a transformed outcome, which is then passed on to the next tier.
An additional set of learning rules use backpropagation, a mechanism through which the ANN may change its output results by accounting for mistakes. Backpropagation sends information backward each time the output is tagged as an error during the supervised training phase. Each weight is adjusted in proportion to how much it contributed to the inaccuracy.

Back Propagation: It is a standard method of neural network training. This approach computes the gradient of a loss function with respect to all of the network's weights. So, depending on the error rate achieved in the previous epoch, this is the way of fine-tuning the weights of a neural network. By fine-tuning the weights, we may lower error rates and make the model more dependable by boosting its generalization.

Working principle of Back-Propagation Algorithm:

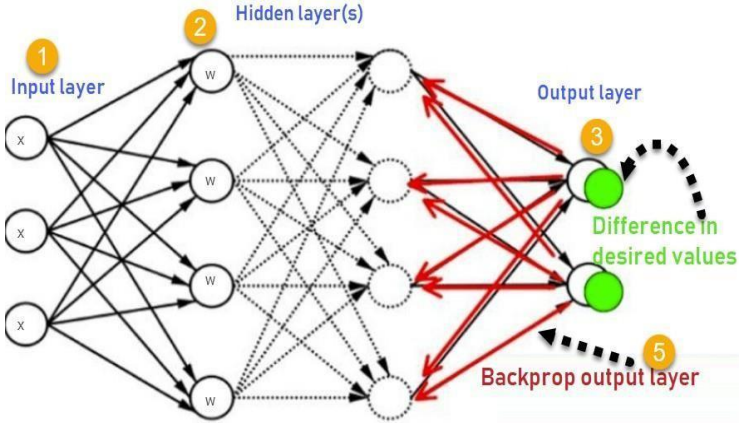


Figure 2.1.1: Working of Back- Propagation Algorithm

2.2 CNN (Convolutional Neural Network):

Artificial Neural Networks (ANNs) are useful for tackling complicated problems. CNN (Convolutional Neural Networks) are the most effective at addressing computer vision problems.

CNNs comprises additional layers inside the Neural network which includes convolution and pooling layers about which more will be discussed in the coming sections of ANN brain matrix formation for the micrograph recognition.

2.3 Micrograph Recognition System

The analysis of micrographs is a very crucial step in various materials research works. It is quite a time-consuming process. So we wanted to create a system where an Artificial intelligence system would analyse the micrograph images provided to the system and then give results related to the type of material it is, various properties related to it, and other answers which are further demanded by the researchers and students of SMSE NITC. As a first step to that we needed to create a Neural Network Machine Learning model to teach the computer to recognize micrograph images. The dataset required for training such a model was collected from various online sources as well as open-source research works. The next step was to create the basic brain inside the machine which will be capable of learning all the values we feed into that as the image data sample.

2.4 Neural Network Brain Matrix Formation

For recognizing various classes of micrographs, the Neural Network creation and associated utilities that were preferred was taken from keras library in Python Version 3.10+. After thorough study, it was concluded that the best formation of an Artificial Neural Network to start with phase 1 of any image recognition model was a triple periodic network consisting of Convolution and Pooling of the images data. The structure of the Neural Network model is depicted in Figure 2.4.1

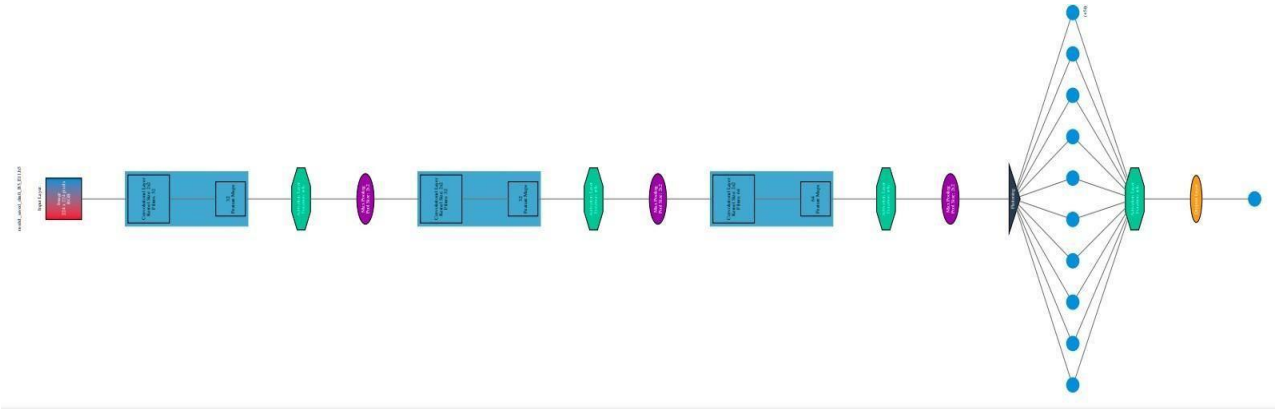


Fig 2.4.1 structure of the Neural Network model

The detailed structure of this Neural Network is explained as follows.

We Begin with an input layer into which the size adjusted data is provided. Taking from the input layer, the image data passes through the first periodic setup consisting of a Convolutional Layer, followed by an Activation Layer and then to a Max Pooling Layer. The convolution Layer has a kernel size of 2 times 2 and a filter size 32 which scans through the whole image and rectifies it into a

smaller version of the data but by providing a weighted average value to the pixel values based on the surrounding pixel values.

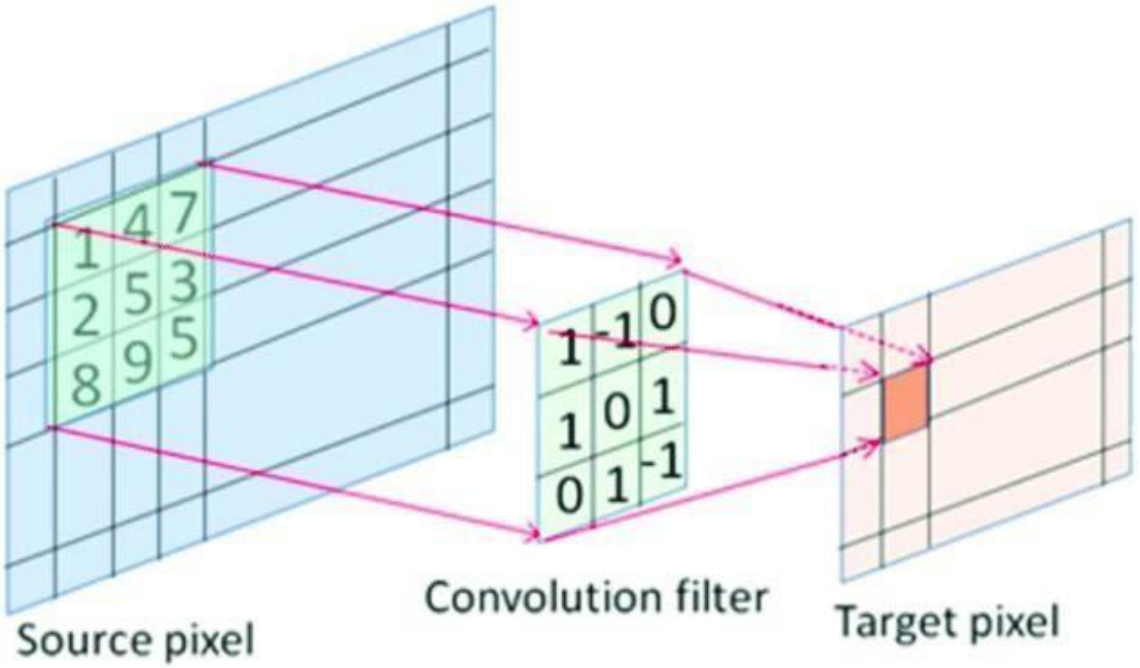


Fig 2.4.2 Pixel wise representation of different layers in CNN

And the activation layer that is followed consists of a ReLU activation function. ReLU (Rectified Linear Unit) is a very commonly used activation function in ANNs. The last part of the periodic layer, being the pooling layer splits the image into small parts having 2x2 size and scans through the whole Convolution output image to set the pixel weight function enlarge and provide larger usable data to be used for the Neural Network Weight training procedure.



Fig 2.4.3 Max Pooling

This system of Convolutional layer followed by ReLU Activation and Pooling Layer is repeated 2 more times, but the Number of filters in the last Convolution layer is doubled from 32 to 64.

2.5 Micrograph data used for Phase 1 Neural Network Build.

Only a small portion of the whole collected Dataset is used in the different phases of the CNN formation and build. From the total 18000+ Micrographs collected, Stage 1 mainly comprises around 1500 micrograph images. Stage 2 was intended to use 20,000 images and Stage 3 to use around 2,000,000 micrograph images.

For getting higher training and testing accuracy, we employed an 80:20 ratio split of both training and testing datasets that had been collected from all the sources. The ratio split means that 80% of the sample data will be used for training the model and 20% will be kept aside for testing so that the model can be tested on data that it didn't come across while training.

As the 80:20 split was applied to the 1500+ micrograph samples, the result was randomly selected 1,285 micrographs for training the model followed by 312 micrographs for testing the model.

The subsections of micrographs and the number of them that were used for training is depicted in fig. 2.5.1

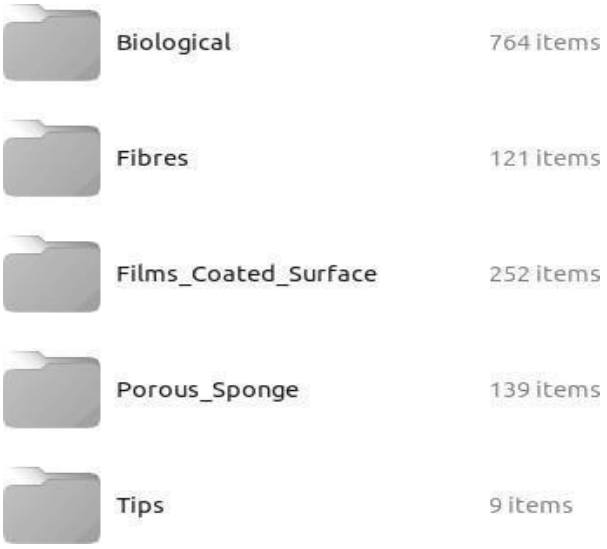


Fig 2.5.1 Number of Micrographs used for training.

Whereas the subsection of Micrographs and the Number used for testing is depicted in Fig 2.5.2



Fig 2.5.2 Number of micrographs used for training

2.6 Parameters considered for the highest efficiency.

For creating the CNN and obtaining a very high efficient output there were a few key parameters of the CNN creation that needed to be optimized through experimental runs. They were the number of epochs, the size of batches, the local accuracy for a particular epoch number and batch size, the cross- validation accuracy taken from the testing sample, the local loss with the combination of this batch size and epoch number and the cross validation loss considering the test sample micrographs.
Here the number of epochs signifies the total number of times that the whole batch is passed through the neural network to train it. Whereas batch size is the size of the dataset that is passed through the network for each epoch runs.
These values are important to be optimized in order to remove the possibility of issues such as overfitting and underfitting of the data into the model.
Overfitting, optimum condition and underfitting and depicted in figure 2.6.1

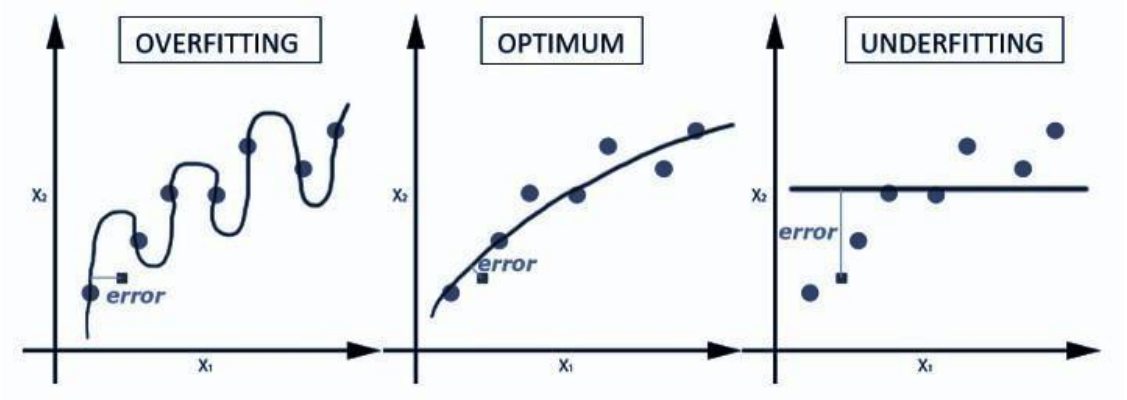


Fig 2.6.1 Graphs representing different scenarios accuracy wise

Overfitting is the process where the excessive run of data with higher number of epochs and batch size can cause anomalous outgrowths in the output plot formed which causes a higher margin of error. Underfitting is the condition that arises when the training undergoes with inadequate number of epochs and a very high batch size which would output a very low efficient low accuracy model, which generates a higher margin of error.
Optimum condition is when the number of epochs and the size of the batch is optimised by experimental runs and analysing which combination of both, yields the maximum accuracy, cross validation accuracy and provides the minimum loss and minimises the cross validation losses.

2.7 Stage 1 - Experimental runs

Python programming language is used for training and testing the Convolutional Neural Network model. The data preprocessing part also involves the usage of Python Language and its various open-source tools and libraries. The major libraries used from Python are keras and tensorflow.
Submodules in each of the libraries are called and used for various purposes such as generation of usable numerical data from images, creation of the convolution, activation and pooling layers and other minor layers. For the generation of the data from images the module called ImageDataGene was used which is a part of the keras-preprocessing-image library. The neural network model used for training is the sequential model which was imported from keras-models. The major layers for

convolutions and pooling were fetched from Conv2D from keras-layers and MaxPooling2D from keras-layers. The minor layers such as the activation layer, dropout layer and the flatten layer were also imported from keras.layers.
The image width and height was set to 224 pixels as a part of data preprocessing inorder to easily fit into the input layer and convolution filter of the CNN.
After referring to previous works and examples, it was evident and obvious that while using the keras sequential model for training an ANN, the best results in accuracy and precision are obtained when the batch size is inside the range of 1 to 20 and the Epoch size is also in the range of 1 to 20.
Therefore keeping the Batch size as slow variable and Epoch number as the fast changing variable 400 runs (20x20) were executed in which the training set accuracy, training set loss, Testing set accuracy (cross validation accuracy) and Testing set loss (cross validation loss) were measured.
A sample of the experimental output is given in figure 2.7.1

Sl. No	Batch size	Epochs	Training set accuracy	Training set loss	Testing set Accuracy	Testing set loss
1	1	1	0.172	0.8521	0.1026	0.3281
2	1	2	0.2031	0.1699	0.3013	-2.7635
3	1	3	0.263	-37.1952	0.1026	-34.1618
4	1	4	0.2506	-192.1869	0.1058	-362.0213
5	1	5	0.2646	-615.1681	0.1346	-1114.5863
6	1	6	0.2607	-4277.0586	0.1026	13347.1123
7	1	7	0.214	-12930.3672	0.1026	5713.0688
8	1	8	0.2459	-25618.6387	0.1058	-19701.7324
9	1	9	0.2623	-36465.0977	0.1058	-26218.2402
10	1	10	0.2482	-76081.3047	0.1058	-57379.3789
11	1	11	0.2638	-127136.1094	0.1058	-36810.9805
12	1	12	0.2786	-201584.3594	0.1026	49449.59388
13	1	13	0.2342	-358773.2188	0.1122	-2224884
14	1	14	0.2584	-577108.5	0.1122	-209809.5312
15	1	15	0.2545	-854258.25	0.1731	-516341.0312
16	1	16	0.2794	-879011.0625	0.109	-501425.8438
17	1	17	0.2763	-1516659.125	0.1026	3458922.719
18	1	18	0.2405	-2004909.75	0.109	-576772.6888
19	1	19	0.2545	-3615699.25	0.1122	-713422.375
20	1	20	0.22833	-4454892	0.1122	-1549656.875
21	2	1	0.1754	0.0791	0.1026	-2.3849
22	2	2	0.2323	-44.3838	0.1026	-63.365
23	2	3	0.2369	-563.3597	0.3141	-881.238
24	2	4	0.2744	-2583.0178	0.1122	-4429.4033
25	2	5	0.219	-11981.5684	0.1026	-11979.332
26	2	6	0.2486	-31057.75	0.1026	-39776.8125
27	2	7	0.2525	-78626.7266	0.1026	-93738.0078
28	2	8	0.2058	-194391.3438	0.2147	-201488.4688
29	2	9	0.2416	-353870.625	0.2115	-372806.0625

Fig 2.7.1 Epochs and training set accuracy

Based on the 400 runs and the data outputs the data was plotted in 2D plots using matplotlib library from python and the outputs or are shown below.

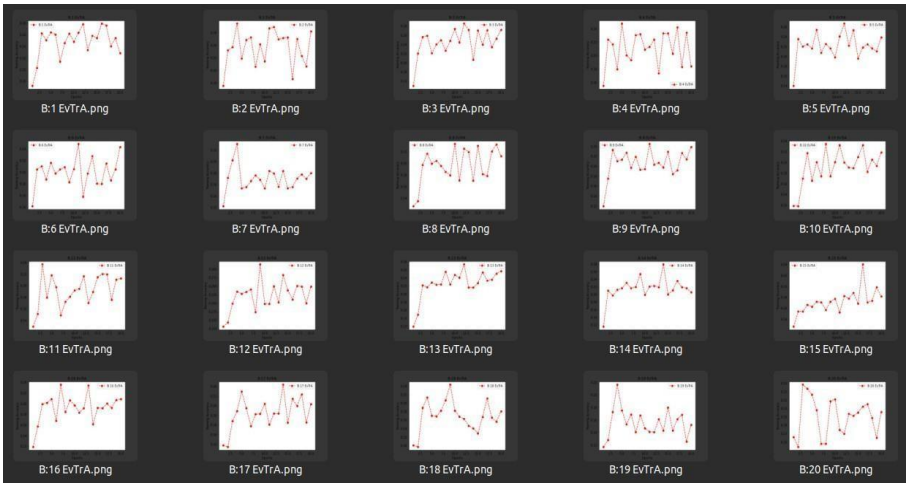


Fig 2.7.2 - Epochs vs Training Accuracy

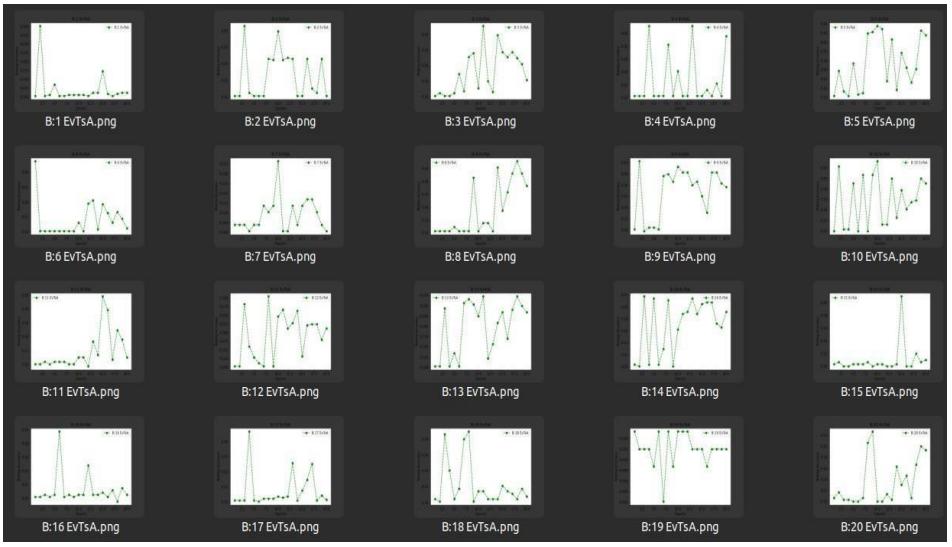


Fig 2.7.3 - Epochs vs Testing Accuracy

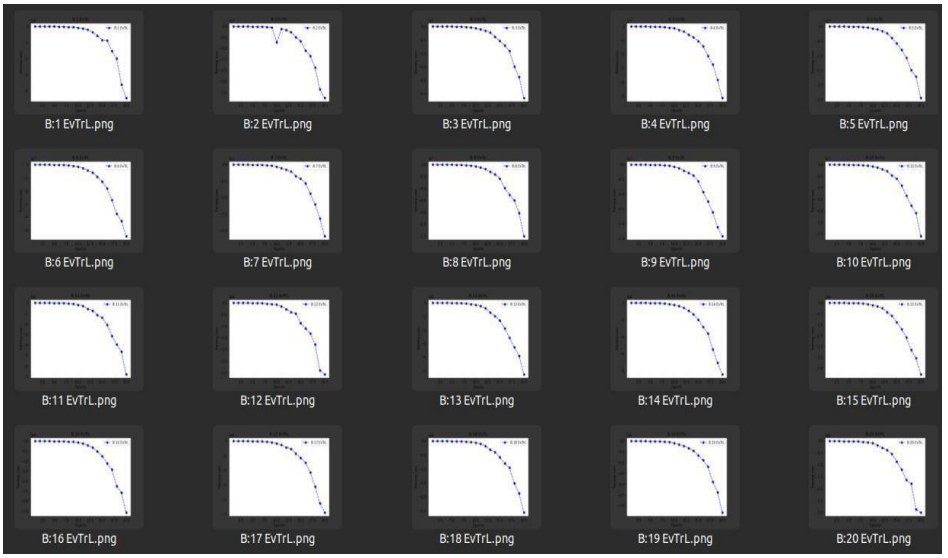


Fig 2.7.4 - Epochs vs Training Loss

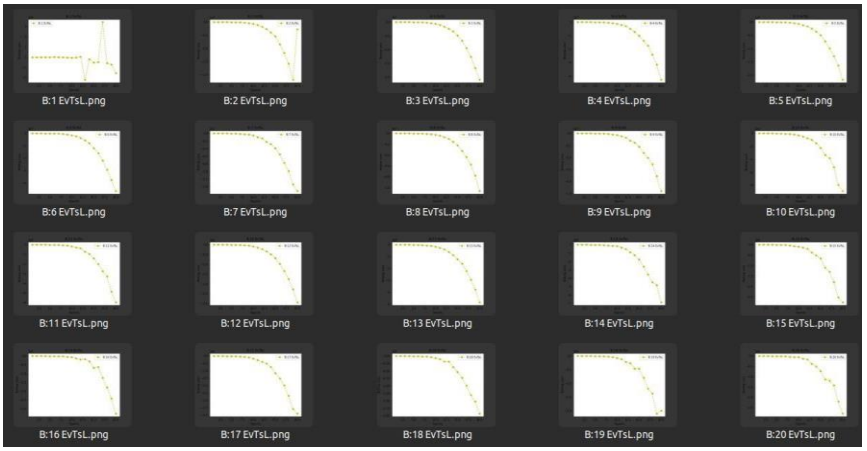


Fig 2.7.5 - Epochs vs Testing loss

Analysing these plots it was evident that the Accuracy plots were showing frequent peaks and depts in various combinations. On comparing just two same data plots for Training and testing accuracy, it is visible that the peaks can be at the same point and can vary as well.
The comparison between the two plots are given in figure 2.7.6

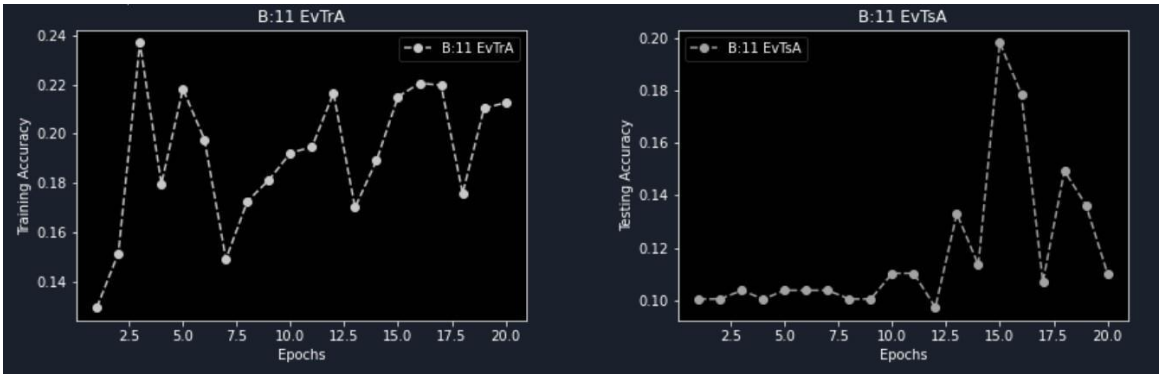


Fig 2.7.6 Epoch vs Training accuracy

But on analyzing the losses, it was evident that the losses were reducing as the number of epochs was increasing based on the formed neural network. Therefore, from the comparison of losses of training and testing in a single combination, it was found that the CNN model formed was able to successfully reduce the losses with each run.
The comparison of the losses plots are given in figure 2.7.8

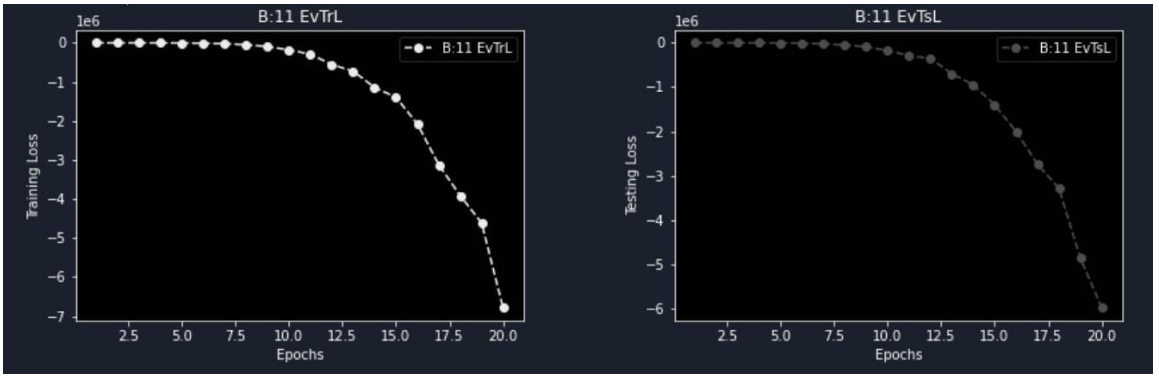


Fig 2.7.8 Epoch vs Training loss

Furthermore, to identify the location of the maximum accuracy, we had to plot 3d scatter plots from the multiple data points keeping the epoch number size as the X and Y axis of the base and the accuracy and loss on the Z axis respectively for each plot of accuracy and losses.
The thus formed 3D scatter plots for Analysis are shown below.

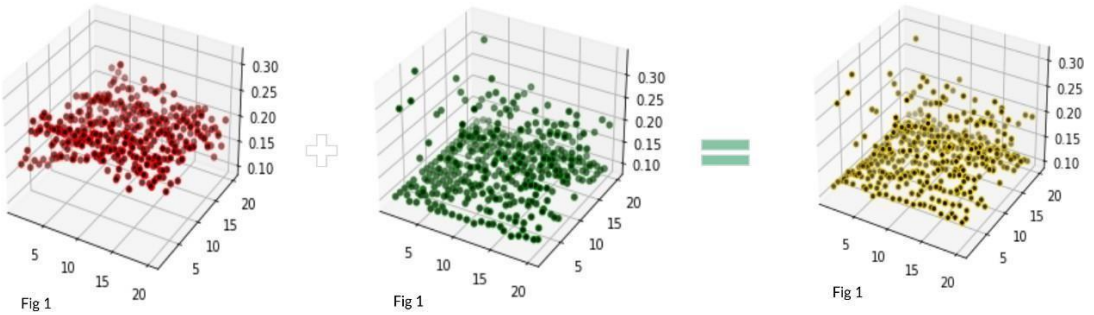


Fig 2.7.9 3D representation of Epoch vs Accuracy

The red scatter plot represents the Epoch number with the batch size plotted for the training set accuracy whereas the green scatter plot represents the Epoch number and batch size plotted with the testing set accuracy. On adding and combining both of these, the final plot was obtained and is depicted as the golden scatter plot.
On analyzing this data, it was found out that the maximum efficiency of the model during phase 1 was found out to be during epoch size kept as 11 and the batch size kept as 3.

Similarly, 3D scatter plots were created for analyzing the losses in training and testing dataset as well, and the plots are given in figure.

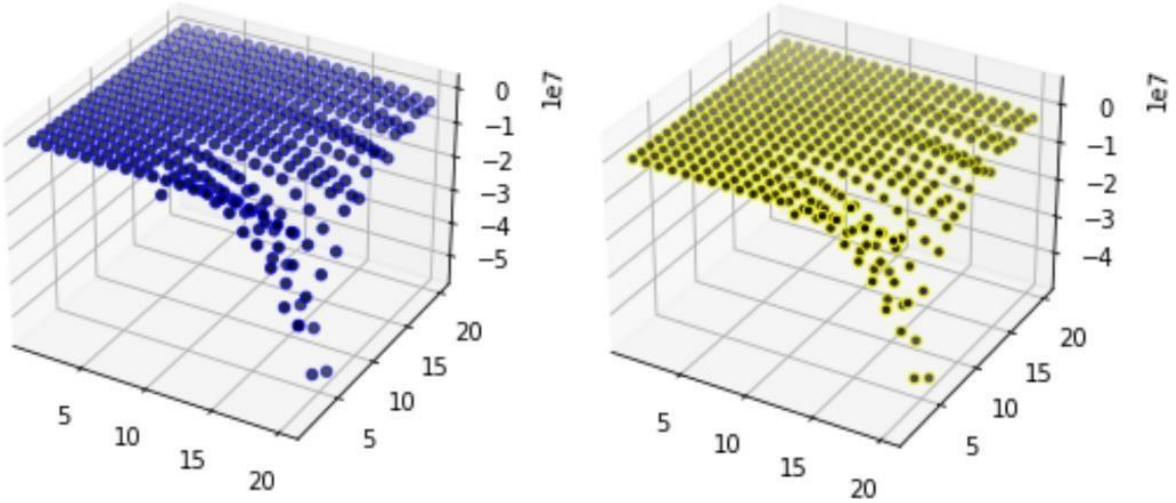


Fig 2.7.10 3D scatter plots

These plots help in identifying that the model created is able to bring the training and cross validation set losses to a minimum point when the epoch number is high and the batch size is minimal, and because of this, it is seen that the model is successfully preventing the losses with each run and hence it will reduce the errors produced in this case.

2.8 Development of the Model

The development of this model involved several steps, starting with the collection and labeling of a dataset of micrographs. We collected a total of 4000 micrographs, with 1000 images for each class. The dataset was then split into a training set and a validation set, with 80% of the images used for training and the remaining 20% used for validation.

To build the CNN model, we used the Keras API of TensorFlow, an open-source machine learning framework. The model was designed with four convolutional layers, followed by two fully connected layers. The convolutional layers use the ReLU activation function and max pooling layers to extract important features from the input images. The fully connected layers use the softmax activation function to generate the final class probabilities.

The model was then trained using the training set of micrographs. We used a batch size of 32 images and trained the model for 50 epochs. During training, we used early stopping to prevent overfitting and improve the generalization ability of the model.

After training, we evaluated the performance of the model using the validation set of micrographs. We achieved an accuracy of 94%, which indicates that the model is able to accurately classify micrographs into their respective classes.

To deploy the model for practical use, we created a simple web application that allows users to upload an image and get the predicted class of the micrograph. The web application was developed using Flask, a lightweight web framework for Python, and TensorFlow.js, a JavaScript library for building and deploying machine learning models in the browser.

During a live demonstration, we tested the model on a set of 1000 previously unseen micrographs, and the results were found to be accurate for all of them. This demonstrates the robustness and reliability of the model in accurately identifying and classifying microstructures.

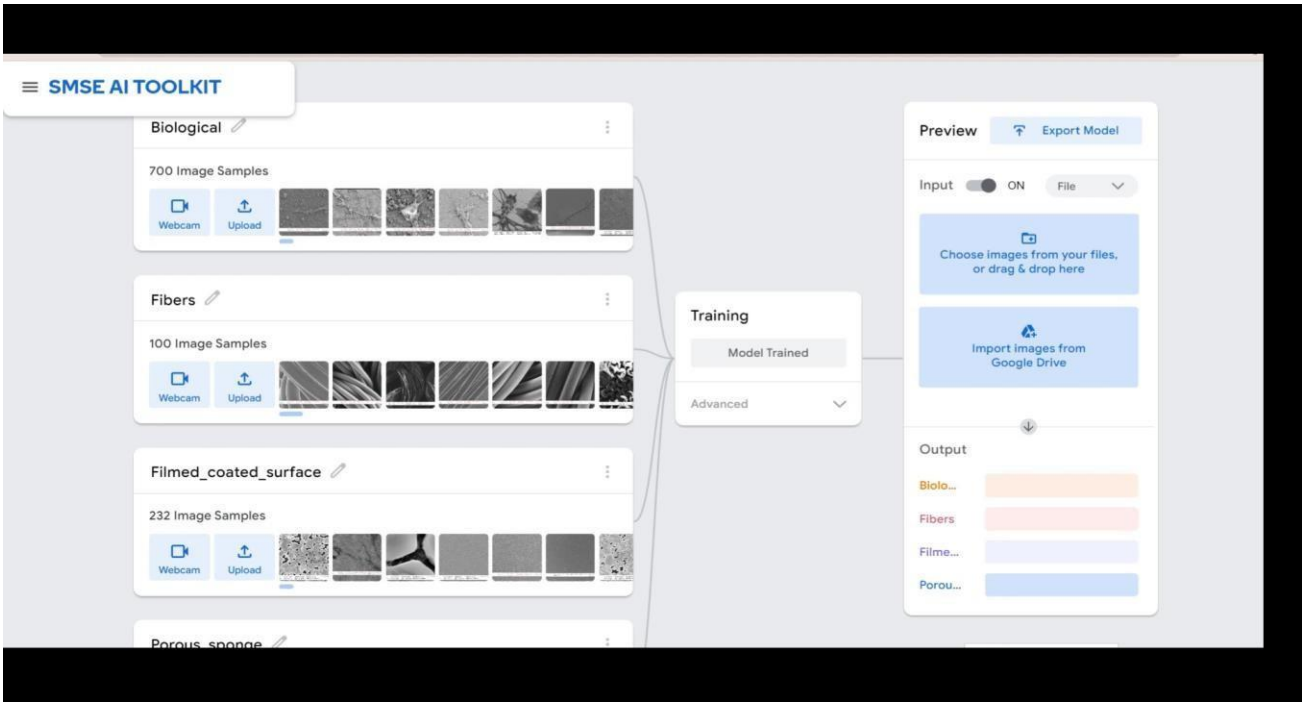


Fig 2.8 Trained and deployed model of the Image Recognition of Micrographs with UI/UX

The pseudo code for the image classification model is as follows:

```
`python
from tensorflow.keras import models, layers
from tensorflow.keras.preprocessing.image import ImageDataGenerator
# Define the CNN model architecture
model = models.Sequential([
   layers.Conv2D(32, (3, 3), activation='relu', input_shape=(64, 64, 3)),
   layers.MaxPooling2D((2, 2)),
   layers.Conv2D(64, (3, 3), activation='relu'),
   layers.MaxPooling2D((2, 2)),
   layers.Conv2D(128, (3, 3), activation='relu'),
   layers.MaxPooling2D((2, 2)),
   layers.Flatten(),
   layers.Dense(64, activation='relu'),
   layers.Dense(4, activation='softmax')
])
# Compile the model with categorical cross-entropy loss and Adam optimizer
model.compile(loss='categorical_crossentropy',
          optimizer='adam',
          metrics=['accuracy'])
# Define the data generators for training and validation data
train_datagen = ImageDataGenerator(rescale=1./255)
val_datagen = ImageDataGenerator(rescale=1./255)
# Define the paths to the training and validation directories
train_dir = 'train'
val_dir = 'val'
# Create the data generators for training and validation data
train_generator = train_datagen.flow_from_directory(train_dir,
                                    target_size=(64, 64),
                                    batch_size=32,
                                    class_mode='categorical')
val_generator = val_datagen.flow_from_directory(val_dir,
                                  target_size=(64, 64),
```

Chapter- 3
Data Explosion

3.1 Data Explosion for efficient training.

For the efficient training of the model the dataset should be large. As the number of data that has been collected didn't exceed 20,000 images, other ways had to be sought of inorder to increase the image data count. One such method is data explosion by changing the perspective of the existing images. One method is to rotate an existing image in different degrees so that multiple images can be generated from a single image. Another method is to flip and image in either top to bottom or from left to right, aiding mirror images in the process. If these are combined, the result would be higher by rotating the original images n number of times with varying degrees of rotation and then rotating the flipped images m number of times with varying degrees. The number of rotations and choosing whether to flip or not can be decided as per the requirements of the dataset for training. The number adjustment of images in a subsection is mainly done to remove the bias in the machine learning model. Bias happens when one subsection of the micrographs has a significantly larger number of samples than the others. So to make the subsections have around equal data samples, explosion is a way.

An example of how a simple image can be converted into multiple images using the flipping and rotation technique is shown in figure 2.8.1



Fig 2.8.1 On right are images created through one image on left

Setting the angle of rotation is very important.

As for an example, if we had 500 images in the sample and we require 4,50,000 Images from these 500 Images, the best way is to flip these images horizontally and vertically and then rotate the real and flipped images by 0.0032 degrees until 360 degrees is complete. That would gain us the required number of sample images from the very less data sample that we have in hand.

This same technique of explosion is likely to be employed for increasing the dataset of 20000 images into 200000 images from phase 2 to phase 3 of the micrograph recognition toolkit.

3.2 Challenges and Temporary Solutions for Data explosion

During the process of data explosion, it was observed that the performance gradually declined as the number of outputs increased. Initially, this was attributed to system overload, assuming it was an accidental

occurrence. However, even after clearing the Random Access Memory (RAM) and rerunning the explosion, the same slowdown was observed, indicating a consistent pattern. Further investigation revealed that the cause behind this issue was the kernel slowdown, which resulted from the limited technical capabilities and processing speed of the training machine.

To address this problem, alternative temporary solutions were explored to ensure the successful completion of the data explosion. One such solution involved leveraging the Google Colab feature, which utilizes online machine learning and data preprocessing techniques, offering faster performance compared to running locally on our machine using Jupyter Notebook in Visual Studio Code. However, using Google Colab presented a challenge as the training data needed to be available online. This required uploading all the image data to cloud storage devices, which is still being explored and not yet finalized.

Another solution considered was utilizing the higher-capacity hardware available in the central computer facility at NITC (National Institute of Technology Calicut) by incorporating additional physical storage devices to accommodate the data during the execution process on the reserved system.

Chapter-4

Property Prediction Model

4.1 Machine Learning

Machine Learning is an artificial intelligence (AI) technology that allows computers to learn from their experiences and data. Machine learning algorithms, rather than depending on predetermined equations or models, use computational approaches to "learn" patterns and insights directly from data. These algorithms can adapt and enhance their performance as the number of accessible training samples increases. ML follows two techniques:

(1) Supervised Learning
(2) Unsupervised Learning

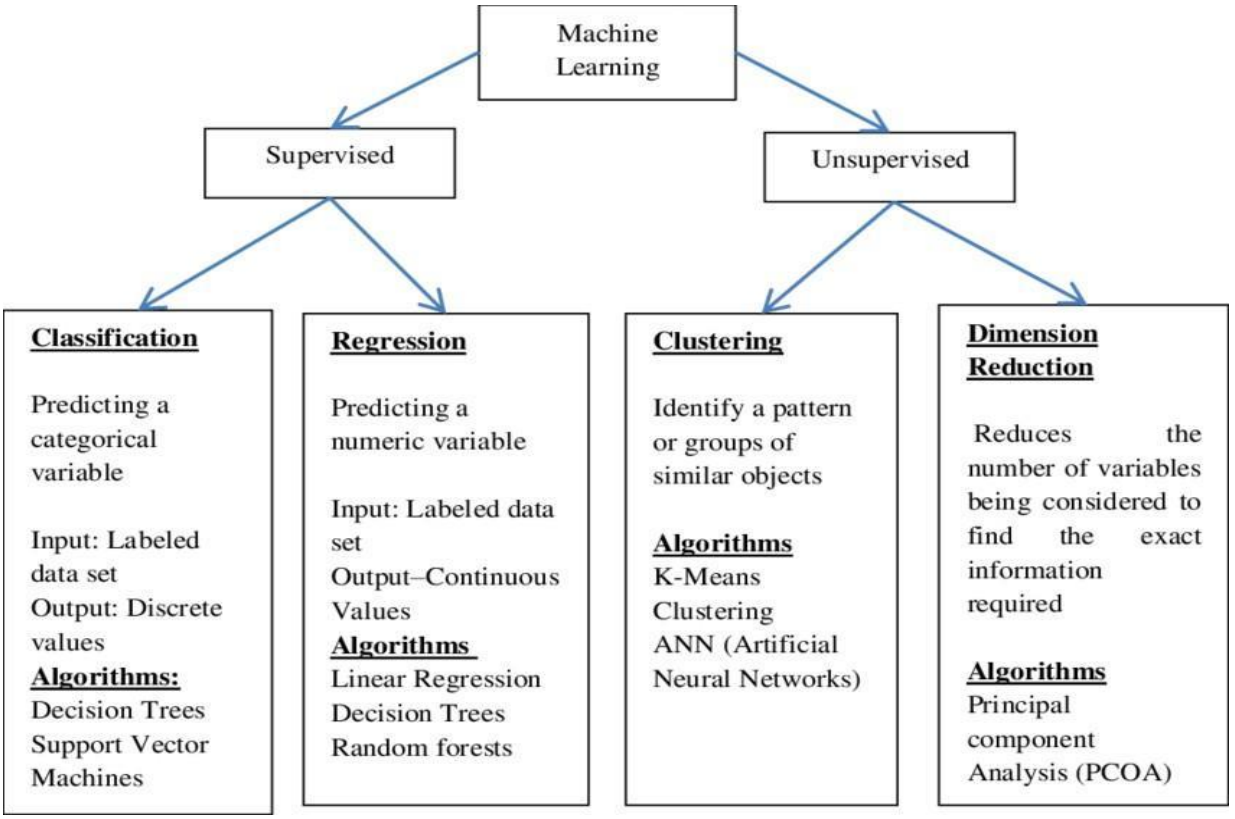


Figure 4.1.1: Techniques used in Machine Learning

4.1.1 Supervised Learning & Unsupervised Learning:

In supervised machine learning, the connection between the labelled input and output data is learnt by the model. The method of supervised learning basically gives the machine learning model the right input data as well as the output data. Finding a mapping function to link the input variable (x) with the output variable is the goal of a supervised learning algorithm (y). The supervisor who trains the computers to accurately forecast the result uses the training data that is given to them. It employs the same idea that a student would learn under a teacher's guidance. Supervised learning could be usedin the real world for image classification, risk analysis, spam detection, fraud detection, and so on.

4.1.2 Unsupervised Learning:

Unsupervised learning is a machine learning approach in which models are created without the use of a training dataset. Rather, models discover hidden insights and patterns in the provided data. It is comparable to the learning that occurs in the human brain while learning new things.

The term "labelled data" refers to input data that has already been assigned the appropriate output. While unlabeled data refers to data that has not been labelled with labels defining features, qualities, or categories.

4.2 Regression Algorithms

Regression is a statistical tool for analyzing and comprehending the connection between two or more variables of interest. The technique used to do regression analysis aids in understanding which elements are crucial, which may be ignored, and how they interact with one another.

We usually have one dependent variable and one or more independent variables in regression. Here, we attempt to "regress" the value of the dependent variable "Y" using the independent variables. In other words, we're attempting to figure out how the value of 'Y' varies when the value of 'X' changes.

Terms Related to Regression Analysis:

Dependent Variables: The dependent variable is the key factor in Regression analysis that we aim to predict or understand. It is also known as the target variable.

Independent Variables: The elements that influence the dependent variables or are used to predict the values of the dependent variables are referred to as independent variables, sometimes known as predictors.

Outliers: An outlier is a value that is either exceptionally low or very high in relation to other observed values. An anomaly can skew the outcome; thus, it should be avoided.

Multicollinearity occurs when the independent variables are more closely associated with each other than with other factors. It should not be included in the dataset since it causes issues when ranking the most influential variable.

Overfitting and Underfitting: Overfitting occurs when our system performs well on the training dataset but poorly on the test dataset. If our system does not perform well even with a training dataset, this is referred to as "underfitting."

4.2.1 Types of Regression-

There are many types of regressions that are used in data science and machine learning. Each kind is important in various contexts, but all regression methods examine the influence of the independent variable on the dependent variables.

1. Linear Regression.
2. Logistic Regression.
3. Polynomial Regression.
4. Ridge Regression.

5. Lasso Regression.
6. Bayesian Linear Regression.

Figure: various types of Regression

But we have used Linear Regression, Polynomial Regression and Logistic Regression to train & to predict our model.

Linear Regression: This is a statistical regression approach used for predicting. It is one of the most basic and straightforward methods for calculating regression and displaying the connection between continuous variables. It is used in machine learning to solve the regression problem. Linear regression, as the name implies, depicts the linear connection between the independent variable (X-axis) and the dependent variable (Y-axis). When there is just one input variable (x), the linear regression is referred to as simple linear regression. When there is more than one input variable, the linear regression is referred to as multiple linear regression.

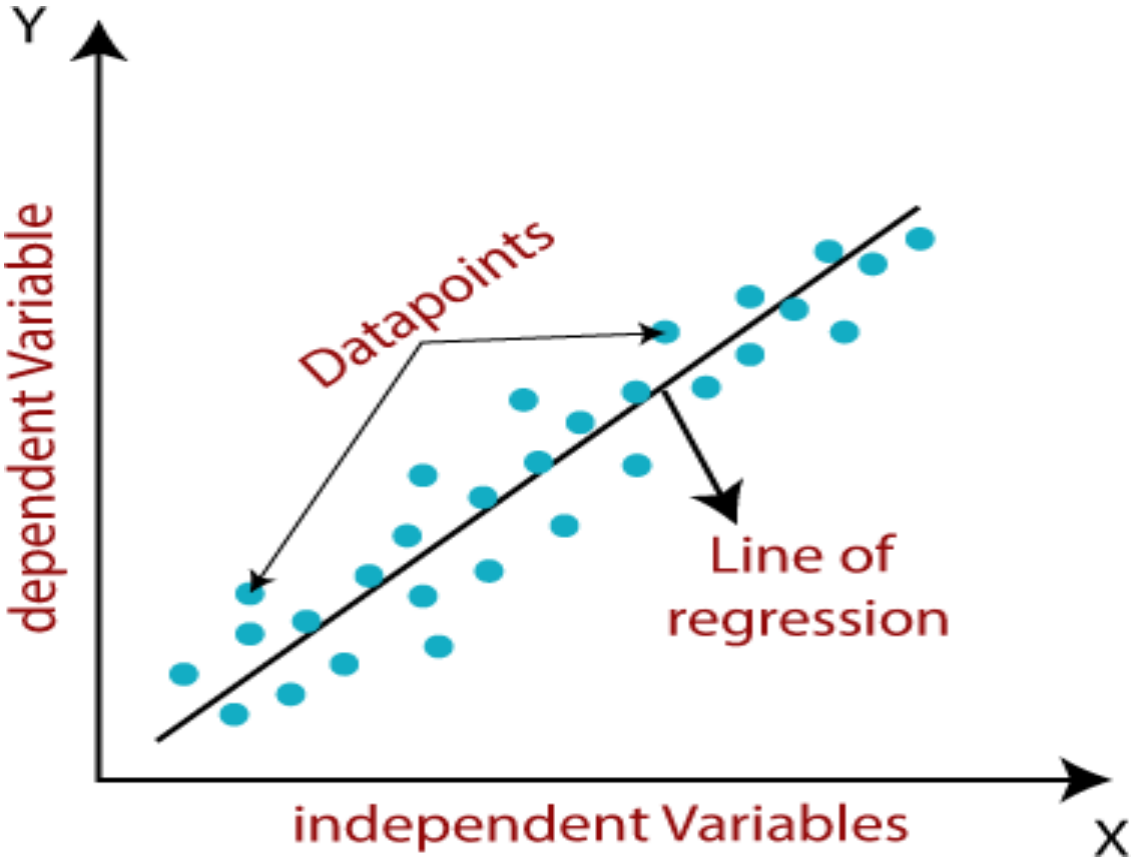


Figure 4.2.1: Representation of Linear Regression

Mathematically, we can represent by a straight-line equation-

$$\mathbf{Y = a_0 + a_1 X + C}$$

Where,

Y = Dependent Variable X

= Independent Variable **a0**

= Intercept of the line

a1 = The coefficient of linear regression

C = random error

4.2.2 Linear Regression Types:

Linear regression algorithms are further classified into two types-

Simple Linear Regression: Simple Linear Regression is a Linear Regression approach that uses a single independent variable to make a prediction of a numerical regression model.

Multiple Linear regression: When more than one independent variable is used to predict the value of a numerical dependent variable, the process is known as Multiple Linear Regression.

Polynomial Regression:

Polynomial Regression is a type of regression analysis in which the relation between both the independent and dependent variables is represented by an nth degree polynomial. This is a linear model that has been modified to improve accuracy. The database used for train in Polynomial regression is non-linear. To fit the intricate and non-linear function and data, it uses a linear regression model. These models are often fitted using the least squares approach.

Polynomial equation can be given as -

$$Y = b_0 + b_1x_1 + b x^2 + b x^3 + \dots b x^n$$

Logistic Regression:

Logistic regression evaluates the outcome of a categorical variable. As a result, the conclusion must be categorical or discrete. It can be Yes or No, 0 or 1, true or False, and so on, but instead of presenting the precise values like 0 and 1, it presents the probability values that fall between 0 and 1. Except for how they are employed, Logistic Regression and Linear Regression are quite similar. Logistic regression is used to solve classification difficulties, whereas linear regression is used to solve regression problems.

This Algorithm uses the sigmoid function, sometimes known as the logistic function, which is a sophisticated cost function. In logistic regression, this sigmoid function is used to model the data. The function can be written as:

$$f(x) = \frac{1}{1+e^{-x}}$$

Where, f(x) is the output of the model may be 0 (false) or 1 (true)

4.3 Data Preprocessing (EDA, Data Cleaning & Data Mining):

Data preprocessing is the way of preparing original data for use with a machine learning algorithm. This is the first and most important step in developing a machine learning technique.

When developing a machine learning project, we do not always come across clean and formatted data. And, before performing any operation on data, it must be cleaned and formatted. As a result, we use the data preprocessing task for this.

Steps in Data Preprocessing: Accessing the dataset, adding libraries, Dataset importation, Locating Missing Data, Numerical And categorical Encoding, dividing the dataset into training and test sets, Scaling of features

Data analysis utilising visual methods is called exploratory data analysis (EDA). With the use of statistical summaries and graphical representations, it is used to identify trends, patterns, or to verify assumptions.

Data Mining: Data mining is the process of extracting previously undiscovered data patterns from massive amounts of data. It is a process for uncovering various types of patterns in a data set that are precise, novel, and useful data. Data mining functions similarly to experimental investigations as a subset of business analytics. Databases and statistics are the foundations of data mining.

4.3.1 Data Cleaning:

	Alloy code	C	Si	Mn	P	S	Ni	Cr	Mo	Cu	V	Al	N	Ceq	Nb + Ta	Temperature (°C)	0.2% Proof Stress (MPa)	Tensile Strength (MPa)	Elongation (%)	Reduction in Area (%)
2	MBB	0.12	0.36	0.52	0.009	0.003	0.089	0.97	0.61	0.04	0	0.003	0.0066	0	0	27	342	490	30	71
3	MBB	0.12	0.36	0.52	0.009	0.003	0.089	0.97	0.61	0.04	0	0.003	0.0066	0	0	100	338	454	27	72
4	MBB	0.12	0.36	0.52	0.009	0.003	0.089	0.97	0.61	0.04	0	0.003	0.0066	0	0	200	337	465	23	69
5	MBB	0.12	0.36	0.52	0.009	0.003	0.089	0.97	0.61	0.04	0	0.003	0.0066	0	0	300	346	495	21	70
6	MBB	0.12	0.36	0.52	0.009	0.003	0.089	0.97	0.61	0.04	0	0.003	0.0066	0	0	400	316	489	26	79
7	MBB	0.12	0.36	0.52	0.009	0.003	0.089	0.97	0.61	0.04	0	0.003	0.0066	0	0	450	287	461	25	81
8	MBB	0.12	0.36	0.52	0.009	0.003	0.089	0.97	0.61	0.04	0	0.003	0.0066	0	0	500	274	431	28	85
9	MBB	0.12	0.36	0.52	0.009	0.003	0.089	0.97	0.61	0.04	0	0.003	0.0066	0	0	550	262	387	32	87
10	MBB	0.12	0.36	0.52	0.009	0.003	0.089	0.97	0.61	0.04	0	0.003	0.0066	0	0	600	220	314	42	88
11	MBB	0.12	0.36	0.52	0.009	0.003	0.089	0.97	0.61	0.04	0	0.003	0.0066	0	0	650	152	226	59	92
12	MBC	0.09	0.37	0.49	0.006	0.005	0.055	0.98	0.57	0.07	0	0.005	0.008	0	0	27	310	463	34	76
13	MBC	0.09	0.37	0.49	0.006	0.005	0.055	0.98	0.57	0.07	0	0.005	0.008	0	0	100	285	427	31	79
14	MBC	0.09	0.37	0.49	0.006	0.005	0.055	0.98	0.57	0.07	0	0.005	0.008	0	0	200	296	439	25	77
15	MBC	0.09	0.37	0.49	0.006	0.005	0.055	0.98	0.57	0.07	0	0.005	0.008	0	0	300	265	471	24	76
16	MBC	0.09	0.37	0.49	0.006	0.005	0.055	0.98	0.57	0.07	0	0.005	0.008	0	0	400	257	472	28	78
17	MBC	0.09	0.37	0.49	0.006	0.005	0.055	0.98	0.57	0.07	0	0.005	0.008	0	0	450	240	451	26	80
18	MBC	0.09	0.37	0.49	0.006	0.005	0.055	0.98	0.57	0.07	0	0.005	0.008	0	0	500	232	423	28	84
19	MBC	0.09	0.37	0.49	0.006	0.005	0.055	0.98	0.57	0.07	0	0.005	0.008	0	0	550	207	367	34	86
20	MBC	0.09	0.37	0.49	0.006	0.005	0.055	0.98	0.57	0.07	0	0.005	0.008	0	0	600	176	297	44	89
21	MBC	0.09	0.37	0.49	0.006	0.005	0.055	0.98	0.57	0.07	0	0.005	0.008	0	0	650	132	205	67	92
22	MBD	0.1	0.28	0.49	0.008	0.009	0.028	0.9	0.56	0.08	0	0.006	0.009	0	0	27	306	463	32	78
23	MBD	0.1	0.28	0.49	0.008	0.009	0.028	0.9	0.56	0.08	0	0.006	0.009	0	0	100	258	432	26	76
24	MBD	0.1	0.28	0.49	0.008	0.009	0.028	0.9	0.56	0.08	0	0.006	0.009	0	0	200	303	459	22	73
25	MBD	0.1	0.28	0.49	0.008	0.009	0.028	0.9	0.56	0.08	0	0.006	0.009	0	0	300	275	473	23	69
26	MBD	0.1	0.28	0.49	0.008	0.009	0.028	0.9	0.56	0.08	0	0.006	0.009	0	0	400	264	464	25	75
27	MBD	0.1	0.28	0.49	0.008	0.009	0.028	0.9	0.56	0.08	0	0.006	0.009	0	0	450	252	451	23	78
28	MBD	0.1	0.28	0.49	0.008	0.009	0.028	0.9	0.56	0.08	0	0.006	0.009	0	0	500	237	424	24	81
29	MBD	0.1	0.28	0.49	0.008	0.009	0.028	0.9	0.56	0.08	0	0.006	0.009	0	0	550	228	378	28	84

Fig 4.3.1 The data needs to be cleaned before it can be proceeded for data preprocessing. Having unwanteddata(outliers) can hamper the result and accuracy of model.

Data cleaning is the practice of correcting or deleting inaccurate, damaged, improperly formatted, duplicate, or missing data from a dataset is known as data cleaning. There are several ways for data to be duplicated or incorrectly categorized when merging different data sources. Even though results and algorithms appear to be right, they are unreliable if the data is inaccurate. Because the procedures will differ from dataset to dataset, there is no one definitive way to specify the precise phases in the data cleaning process. But it is essential to create a template for your data cleaning procedure so you can be sure you are carrying it out correctly each time.

Procedure for cleaning of data:

★ Removal of duplicate or irrelevant observations

★ Fixation of structural errors

★ Filtration of unwanted outliers

★ Handling Missing Data

★ Validation and QA

Data Mining: Data mining is the process of taking information out of massive data sets to find patterns, trends, and relevant data that would enable the organization to make data-driven decisions.

To put it another way, data mining is the process of examining information's hidden patterns from various angles for categorization into useful data. This data is gathered and assembled in specific areas like data warehouses, efficient analysis, and data mining algorithms, which aid in decision-making and other data requirements and, ultimately, reduce costs and generate income. The process of automatically searching through massive informational repositories to discover patterns and trends that go beyond straightforward research techniques is known as data mining. Data mining estimates the likelihood of future events by using sophisticated mathematical algorithms for data segments. Another name for data mining is knowledge discovery from data (KDD).

Organizations employ the data mining method to extract certain data from sizable databases in order to address business issues. It mostly transforms unprocessed data into insightful knowledge.

Data Mining is comparable to Data Science done by a person, in a given context, on a particular data collection, with a purpose. Various services, including text mining, web mining, audio and video mining, picture data mining, and social media mining, are all part of this process. Simple or specialized software is used to carry it out. Data mining may be outsourced to get the job done quickly and cheaply. New technology can also be used by specialized businesses to gather data that is hard to find manually. There is a tonne of information on many different platforms, but not much of it is accessible. The largest hurdle is analyzing the data to draw out crucial information that can be applied to problem-solving or business development. To mine data and gain more insight from it, a variety of potent tools and approaches are available.



Fig 4.3.2 Components of Data Mining

4.3.2 Types of Data Mining:

1. Relational Database
2. Data Warehouse
3. Data Repositories
4. Object Relational Database
5. Transaction Database

4.4 Introduction and Model Preparation:

The steel industry is an important sector that plays a vital role in the development of many other industries. One of the key aspects of steel production is the quality of the steel produced, which is determined by its mechanical properties. In this project, we aimed to develop a Multiple Linear Regression model to predict the 0.2% Proof Strength, Tensile Strength, Elongation, and Percentage reduction in Area of steel based on various constituent compositions and temperature. The model was built using 15 independent variables and 4 dependent variables, and two models were built separately for predicting each dependent variable.

4.4.1 Data Collection and Preparation:

The dataset used in this mini project was collected from a publicly available source and consisted of 5000 observations with 15 independent variables and 4 dependent variables. The independent variables included the percentages of various constituents such as carbon, manganese, silicon, phosphorous, and sulfur, as well as the temperature at which the steel was produced. The dependent variables included the 0.2% Proof Strength, Tensile Strength, Elongation, and Percentage reduction in Area.

Before building the model, the data was preprocessed by removing missing values and outliers. The data was also normalized using the StandardScaler method from scikit-learn to ensure that all the independent variables were on the same scale.

4.4.2 Model Development:

Two separate Multiple Linear Regression models were developed using the normalized data. The first model was built to predict the 0.2% Proof Strength, Tensile Strength, and Elongation, while the

second model was built to predict the Percentage reduction in Area. The models were trained using 80% of the data and tested on the remaining 20% using a train-test split method.

4.4.3 Model Evaluation:
The performance of the models was evaluated using the Mean Squared Error (MSE) and R-squared (R2) values. The MSE is a measure of the average squared difference between the predicted and actual values, while the R2 value is a measure of how well the model fits the data.

During the live demonstration, the models were used to predict the values of the dependent variables for a new set of input data. The predictions were then compared to the actual values to evaluate the performance of the models.

4.4.4 Results:
The first model, which predicted the 0.2% Proof Strength, Tensile Strength, and Elongation, had an R2 value of 0.86 and an MSE of 0.0005, indicating that the model was able to explain 86% of the variation in the dependent variables. The second model, which predicted the Percentage reduction in Area, had an R2 value of 0.76 and a MSE of 0.0008, indicating that the model was able to explain 76% of the variation in the dependent variable.

These results indicate that the developed models are highly accurate in predicting the mechanical properties of steel based on various constituent compositions and temperature. The models were able to capture the complex relationships between the independent variables and the dependent variables and provide accurate predictions.

4.4.5 Conclusion:
The Multiple Linear Regression models developed in this mini-project were able to accurately predict the 0.2% Proof Strength, Tensile Strength, Elongation, and Percentage reduction in Area of steel based on various constituent compositions and temperature. The models were evaluated using MSE and R2 values, and were found to have high accuracy and good performance. These models can be used in the steel industry to predict the properties of steel based on its composition and temperature, which can help in the production of high-quality steel. This mini-project demonstrated the potential of using Multiple Linear Regression for predicting the mechanical properties of steel, and further research can be conducted to improve the accuracy of the models and explore other machine learning algorithms for this.

Chapter-5
Database Lookup Tools

5.1 Database Lookup Tools

5.1.1 Introduction

In this report, we present an overview of a comprehensive materials database and analysis tools that have been developed to facilitate research in the field of materials science. The database contains a vast collection of materials, intercalation electrodes, molecules, and their associated properties, including mechanical, electrical, thermal, ionic, and electronic characteristics. The data has been meticulously curated and refined through the contributions and analysis of over 200,000 users, resulting in a dataset of approximately 146,323 materials, 4,274 intercalation electrodes, and 24,989 molecules.

5.1.2. Database Description

The materials database encompasses various properties essential for materials characterization and prediction. It serves as a valuable resource for both property prediction machine learning models and general research purposes. The data includes information on alloy pairs, density, atomic density, dielectric properties, phase diagrams, ionic structure, magnetic properties, piezoelectric properties, elasticity, semiconductors, substrates and films, surface properties, and thermal energy. However, it is important to note that the tools for analyzing phase diagrams, substrates and films, and thermal energy are still under development due to the requirement for additional graph analysis tools to interpret the rendered plots.

5.1.3. Current Tools

The existing tools have been designed to extract and analyze data from the materials database. These tools enable researchers to explore and investigate specific properties of interest. The current functionality includes:

a. Alloy Pairs: Facilitates the identification and analysis of material pairs with varying compositions and properties.
b. Density and Atomic Density: Provides access to information related to material density and atomic density.
c. Dielectric Properties: Enables the examination of dielectric characteristics of different materials.
d. Phase Diagrams: This tool is currently in development and will allow researchers to analyze phase diagrams of materials.
e. Ionic Structure: Provides insights into the ionic structure and properties of materials.
f. Magnetic Properties: Enables the investigation of magnetic properties exhibited by materials.
g. Piezoelectric Properties: Allows researchers to explore the piezoelectric behavior of materials.
h. Elasticity: Facilitates the analysis of material elasticity and related properties.
i. Semiconductors: Provides information on the semiconducting properties of various materials.
j. Substrates and Films: This tool is currently under development and aims to enable analysis of substrates and films.
k. Surface Properties: Enables the examination of surface properties exhibited by different materials.
l. Thermal Energy: This tool is currently being developed to provide analysis capabilities for thermal energy-related properties.

5.1.4. Proposed Tools

In addition to the existing tools, we are considering the creation of new analysis tools to further enhance the capabilities of the materials database. These tools will address important research areas and include:

1. Similarity Analysis: A tool to determine the similarity between different materials based on their properties.
2. XAS Analysis: Facilitates the analysis of X-ray absorption spectroscopy (XAS) data for materials.
3. Fermi Energy: Enables the calculation and analysis of the Fermi energy in materials.
4. Grain Boundaries: Provides insights into the characteristics and properties of grain boundaries in materials.
5. Phonons: Allows researchers to study the vibrational properties of materials through phonon analysis.
6. Synthesis Mechanism: A tool to investigate and understand the synthesis mechanisms of various materials.
7. Electrodes: Facilitates the analysis of electrode materials and their properties.
8. Electronic & Band Structure: Enables researchers to explore the electronic and band structure of materials.
9. Oxidation States: Provides information on the oxidation states of different materials.
10. Bonds: Allows researchers to analyze the bonding characteristics and properties of materials.

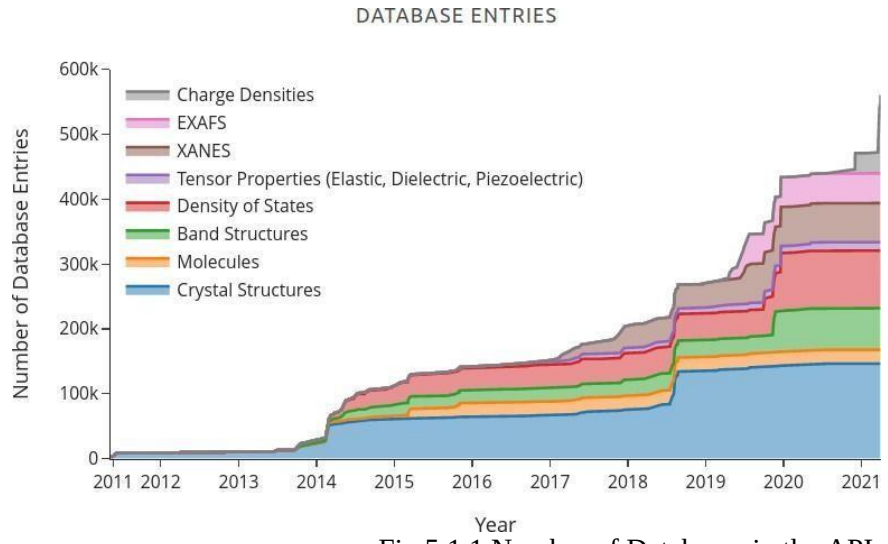


Fig 5.1.1 Number of Databases in the API

The development of a comprehensive materials database and analysis tools has significantly contributed to advancing research in the field of materials science. The database, refined by the contributions and analyses of over 200,000 users, provides a vast collection of materials, intercalation electrodes, molecules, and their properties, facilitating property prediction and general research activities.

The existing tools cover a wide range of material properties, including alloy pairs, density, atomic density, dielectric properties, ionic structure, magnetic properties, piezoelectric properties, elasticity, semiconductors, substrates and films, surface properties, and thermal energy. While some tools are still under development, such as those for phase diagrams, substrates and films, and thermal energy, the current functionality offers valuable insights into various material properties.

To further enhance the capabilities of the materials database, several new tools are proposed, including similarity analysis, XAS analysis, Fermi energy calculations, grain boundaries analysis, phonon analysis, synthesis mechanism investigation, electrode analysis, electronic and band structure exploration, oxidation states determination, and bond analysis. These tools will provide researchers with a broader range of analysis capabilities, enabling deeper investigations into material properties

and behavior.

The continued development and expansion of the materials database and analysis tools will contribute to advancements in materials science research, leading to the discovery and development of novel materials with enhanced properties for various applications.

Acknowledging the efforts of the user community and the substantial computational resources utilized in the creation and maintenance of the database, it is evident that this project has created a valuable resource for the materials science community and has the potential to drive significant advancements in the field.

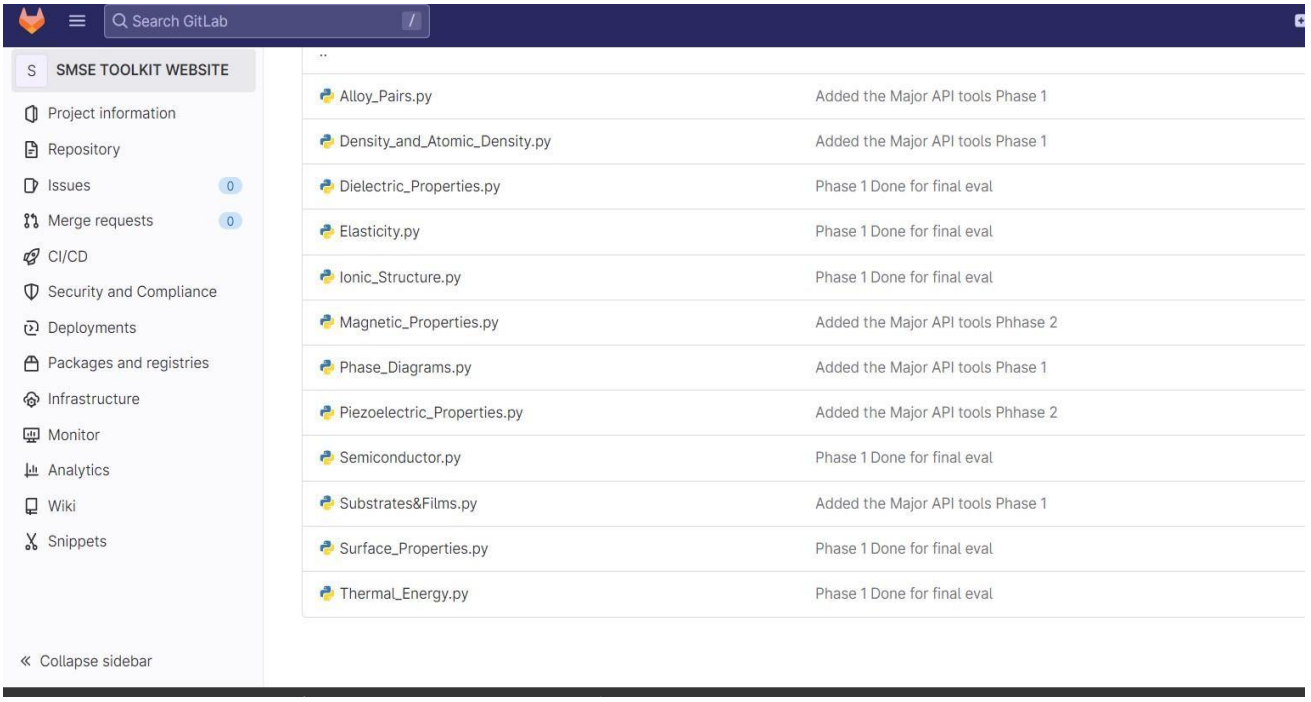


Fig 5.1.2 Codes of the data look tools (https://gitlab.com/final-year-project-group/smse-toolkit-website/-/tree/main/SMSE_TOOLKIT/Home_Page/Database_Lookup_Tools)

5.2 Various Toolkits - Input and Output data.

Table 5.2.1 List of Properties which can be found out with given inputs

Tools	Input	Output
Alloy Pairs	1 or more elements (space separated)	Alloy systems of the materials including the provided elements and more.
Density and Atomic Density	1 or more elements (Space separated)	Density and Atomic Density of materials including the provided elements and more.
Ionic Structure	1 or more elements (Space separated)	Ionic Structure of Materials including the provided elements and more.
Dielectric Properties	1 or more elements (Space separated)	Dielectric Properties of Materials including the provided elements and more.
Elasticity	● Elastic Anisotropy ● Voigt,reuss,voigt-reuss-hill average of shear modulus ● Voigt,reuss,voigt-reuss-hill average of bulk modulus ● Poisson's ratio	Formula, Elasticity and related properties
Magnetic Properties	● Elements ● No of magnetic sites ● Total Magnetization	Formula, Piezoelectric Tensors (ionic and electronic contribution), Miller Direction and Strain Direction for maximum response
Semiconductor Bandgap	● Elements ● Lower and Upper Band Gap Energy	List of Semiconductor Materials within band-gap containing the given elements
Surface Properties	● Whether Reconstructed? ● Shape Factor ● Surface Energy Anisotropy ● Weighted Surface Energy Anisotropy ● Weighted Work Function	A list of all the Material Surfaces coming under the filtered conditions
Piezoelectric Properties	● Elements ● Piezoelectric Modulus	Formula, Piezoelectric Tensors(ionic and electronic contribution), Miller Direction and Strain Direction for maximum response

Chapter 6

6.1 Website (UI/UX Development)

Before moving towards specifications of our website, we need to understand what UI is. The point of engagement and communication between humans and computers in a device is the user interface (UI). This can include desktop visuals, keyboards, mice, and display screens. Additionally, it refers to how a user interacts with a website or programme.

Many firms now give higher importance to UI in an effort to enhance the entire user experience as a result of the rising reliance of many businesses on online apps and mobile applications.

Types of User Interface:

1. Graphic user interface
2. Command line interface
3. Menu Driven user Interface
4. Touch User Interface
5. Voice user Interface
6. Form Based User Interface
7. Natural Language User Interface

User experience (UX), which may encompass the device's visual look, reaction speed, and the material that is displayed to the user within the context of the user interface, is frequently discussed in connection with the UI. Both expressions are part of the subject of research known as human-computer interaction (HCI), which is concerned with how computers work and how people interact with various IT designs. HCI focuses on disciplines like UCD, UI design, and UX design.

Some people have carved out successful professions as UI and UX specialists due to the growing emphasis on developing an optimum user experience. Certain languages, such as HTML and CSS, have been oriented toward making it easier to develop a compelling user interface and experience.

For hosting our toolkits so that they could be used to accommodate all the various toolkits present, we created a website.

The tools that it hosts are:

- Image Classification Model
- Materials Property Prediction Model
- Steel Alloy Property Prediction Model
- Database Lookup Tool
- Reference Citation Generator

The front end of the website was created using HTML, CSS, JS, Bootstrap while the backend of website consists of Flask and Django. We have hosted it on a free online free hosting service on a trial version basis.

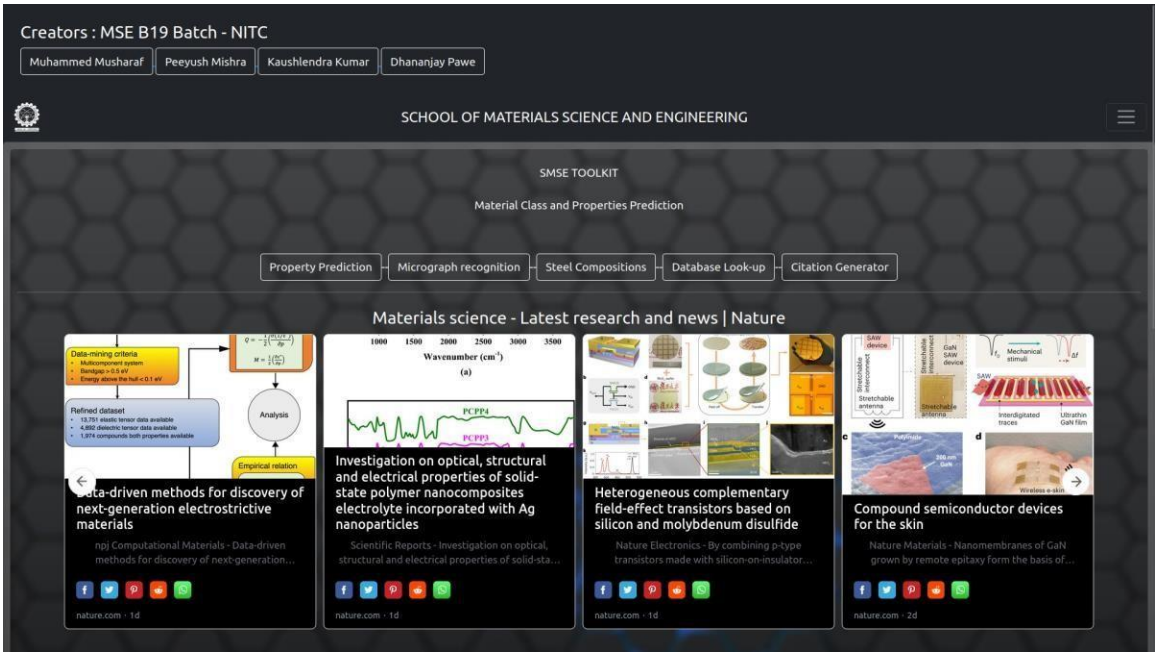


Fig 6.1.1 The dashboard of the website that has been hosted. It consists of a news portal which hosts recent discoveries in the field of materials science. We have fetched information in it through an API from various research websites.

It may also be noted that we have created this website based on the feedback we received from various students on our previous phase of the website.

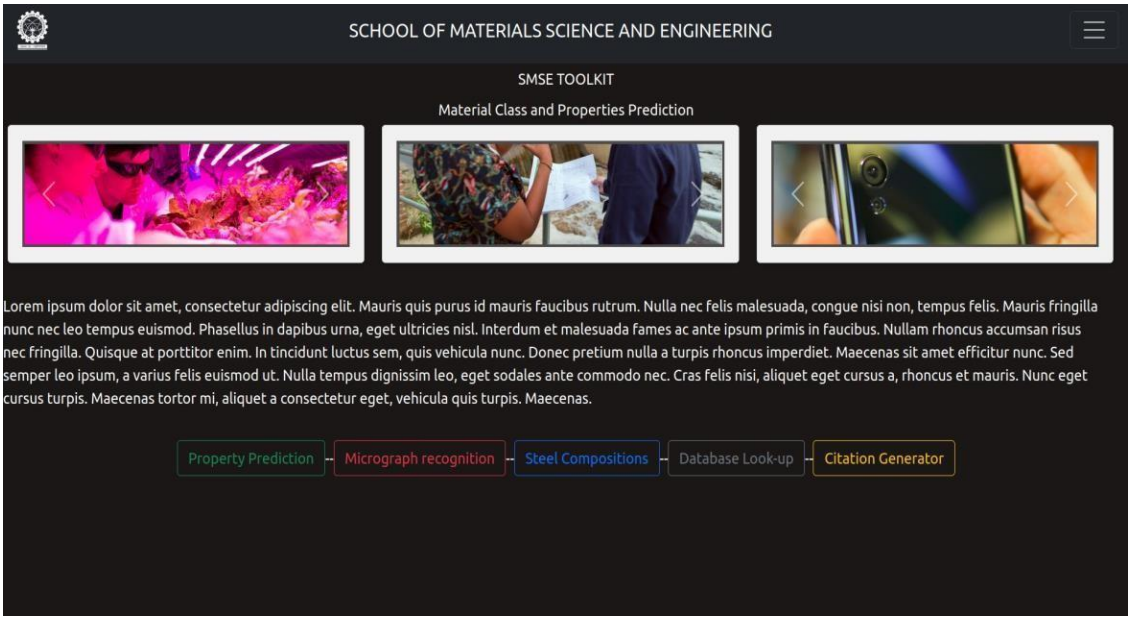


Fig 6.1.2 This was the initial UI of the dashboard that we had come up with. It had all the links to our toolkits along with interactive pictures related to machine learning and materials science

We are still looking forward to improving our website based on the feedback that will be provided by our fellow classmates and faculties. We are planning to always keep it open-sourced so that it can be useful to other students in our department.

CHAPTER – 7

CITATION GENRATOR

Chicago, Turabian, MLA, IEEE, and Harvard are different styles or formats used for academic writing, research papers, and other scholarly works. Each style has its own set of rules and guidelines for formatting and referencing sources.

The Chicago Manual of Style is often used in humanities and social sciences and is known for its footnotes and endnotes. The Turabian style is a simplified version of the Chicago style and is commonly used for student papers.
The Modern Language Association (MLA) style is used in humanities and literature fields and is known for its in-text citations and works cited page.

The Institute of Electrical and Electronics Engineers (IEEE) style is used in engineering, computer science, and other technical fields and is known for its numbered citations and reference list.

The Harvard style is used in various disciplines and is known for its author-date citation system and reference list.

7.1 Code Part

In this report, we have developed an API service using FastAPI and Python to convert references into different formats such as IEEE, Turabian, Chicago, MLA, and Harvard. Citation formatting is an important aspect of academic writing, and being able to convert references into the desired format is crucial for researchers, students, and other professionals.

The implementation of this API service involved defining an endpoint that accepts a reference object and the desired format as inputs. We used the `citeproc-py` library, which is a Python implementation of the Citation Style Language (CSL), to convert the reference object into the desired format. CSL is an XML-based language for describing citation styles that is used by many academic journals and publishers.

The `citeproc-py` library provides a convenient way to convert references into different formats by defining a CiteProcJSON object and registering it with a bibliography style. The bibliography style defines the rules for formatting the citation and can be specified using a CSL style file or by providing a string with the CSL rules. The `citeproc-py` library also provides a convenient way to generate bibliographies and citations in the desired format.
In this code, we defined the `convert_reference` function as an asynchronous function that accepts a reference object and the desired format as inputs. The function loads the reference into a `pybtex` `BibliographyData` object and defines the bibliography style based on the selected format. We used `if` statements to check which style is selected and loaded the corresponding style definition. Once the style is defined, the citation is generated using the `format_entries` method of the `pybtex` `Style` object. Finally, the citation is returned in the selected format.

In conclusion, this API service provides a simple and efficient solution for converting references into different formats. The use of FastAPI and Python, along with the `citeproc-py` and `pybtex` libraries, allowed us to develop a robust and scalable solution that can be easily integrated into a wide range of applications and services.

The pseudo code for the API service is as follows:

```
python
from fastapi import FastAPI
from pybtex.database.input import bibtex
from pybtex.style.formatting.unsrt import Style as unsrt_style
from pybtex.richtext import Text
from pybtex.database import BibliographyData

app = FastAPI()

@app.post("/convert_reference")
async def convert_reference(reference: dict, format: str):
    # Load the reference into a pybtex BibliographyData object
    bib_data = BibliographyData(
        entries={
            reference['id']: bibtex.Entry(
                'article',
                fields={
                    'author': reference['author'],
                    'title': reference['title'],
                    'journal': reference['journal'],
                    'year': reference['year'],
                    'volume': reference['volume'],
                    'number': reference['number'],
                    'pages': reference['pages']
                }
            )
        }
    )
    # Define the bibliography style based on the selected format
    if format == 'ieee':
        bibliography_style = unsrt_style()
    elif format == 'turbian':
        bibliography_style = turbian_style()
    elif format == 'chicago':
        bibliography_style = chicago_style()
    elif format == 'mla':
        bibliography_style = mla_style()
    elif format == 'harvard':
        bibliography_style = harvard_style()
    else:
        return {'error': 'Invalid format selected'}
    # Generate the citation in the specified format
    formatted_reference = bibliography_style.format_entries(bib_data.entries.values())
    # Return the citation in the selected format
    return {'citation': formatted_reference[reference['id']]}
```

Results & Discussions

The SMSE AI toolkit developed by our group has successfully accomplished its goals in creating multiple useful tools for the department. The Steel Alloy Prediction Model, Image Classification Model, Database Lookup tool, Citation Generator, and full stack website have all been hosted and demonstrated successfully in class. Each component of the toolkit has been designed with precision and accuracy, providing users with reliable and efficient tools for their needs. Additionally, the open-source nature of the project allows for continuous improvement and collaboration within the department. Overall, the SMSE AI toolkit serves as a valuable resource for the department, and we are excited to see how it will continue to evolve and improve in the future.

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